IEEE CIRCUITS AND SYSTEMS SOCIETY

UNIC-CASS

Mock Tapeout 2

“250 MHz *Quadrature Delay Locked Loop* ”

27.01.2026

Fundación FULGOR

UNS UNIVERSIDAD NACIONAL DEL SUR

UTN Facultad Regional Córdoba

UNC Universidad Nacional de Córdoba

Content

1. **2nd UNIC-CASS Progress Questionnaire**
2. **QDLL Project:** Block Diagram and Specifications
3. **Schematics and Layout**: TOP and Sub-blocks: Symbols & Schematics & Layouts
4. **QDLL TOP Layout Verification:** Top Layout and layout within the pad ring.
5. **QDLL:** Pins and IO Cells.
6. **Project Status**

2nd UNIC-CASS Progress Questionnaire

Progress Questionarie

Design Team Identifier (Use an ACRONYM as a short name for the IP block top level) *

QDLL

Primary contact email *

vazquezleonardodavid@outlook.com

Secondary contact email *

leonardo.vazquez@ieee.org

Active Design Team members (First Name & Last Name of all participants, except mentor) *

Leonardo David Vazquez
Santiago Basignana
Mateo Buteler
Clemente Molinari

Mentor's Name(s) *

Agustín Galetto

Repository link (e.g. https://github.com/username/repository-name) *

https://github.com/leonardovazquez/unic-cass-wrapper_QDLL
leonardovazquez/unic-cass-wrapper_QDLL: Unic-Cass Repo for the analog project "QDLL" in 130nm sg13g2 tech

Link to tapeout report (provide link to e.g. Google docs file with your steps and outputs) *

https://docs.google.com/presentation/d/1SaUPbj3oBROP9z1-2TtK8tWIxL02-J9qEyfox9xo-il/edit?usp=sharing

https://docs.google.com/presentation/d/1SaUPbj3oBROP9z1-2TtK8tWIxL02-J9qEyfox9xo-il/edit?usp=sharing

Design Team Country *

ARGENTINA

Progress Questionarie

Is your design? *

- ○ Digital Only
- ◉ Analog / Mixed Signal

Do you have special I/O requirements (RF Signal, High Voltage, High Current, low R/C Parasitics, low leakage)?
If so, please explain with as much details as possible

The QDLL project basically consist on a single block with 1 differential input signal (2 inputs) and 1 differential output signal (2 outputs). For a CMOS (0-1.2 V) 250•MHz input signal, standard IO cells have been tested and found to perform well. However, if available, we prefer to interface with RF 50•Ω systems using IO cells with matched impedance. We would need the new RF iocells models. If this is not possible, as we have demonstrated, we will use the IO Cells sg13g2_IOPADAnalog that we mentioned in our document. Also, we can migrate our project to a lower frequency, there is no problem. Thanks!

For Analog / Mixed-Signal Design

Is the circuit schematic for your Design already completed in Xschem? *

- ◉ Yes
- ○ No - Explain below

If the schematic is not completed, tell us where you are at

Tu respuesta

What is your Core Area in mm² (without the pads!) *

- ◉ < 0.1 mm²
- ○ 0.1 mm² to 0.2 mm²
- ○ 0.2 mm² to 0.3 mm²
- ○ 0.3 mm² to 0.4 mm²
- ○ 0.4 mm² to 0.5 mm²
- ○ 0.6 mm² to 0.7 mm²
- ○ 0.7 mm² to 0.8 mm²
- ○ 0.8 mm² to 0.9 mm²
- ○ 0.9 mm² to 1 mm²
- ○ > 1 mm²

Progress Questionarie

What is your Core Area in mm² (without the pads!) *

- (selected) < 0.1 mm²
- 0.1 mm² to 0.2 mm²
- 0.2 mm² to 0.3 mm²
- 0.3 mm² to 0.4 mm²
- 0.4 mm² to 0.5 mm²
- 0.6 mm² to 0.7 mm²
- 0.7 mm² to 0.8 mm²
- 0.8 mm² to 0.9 mm²
- 0.9 mm² to 1 mm²
- > 1 mm²

How many VDD Pads (0-2) *

1

Which voltage option are you using as your supply voltage (VDD): *

- [x] 1.2V
- [ ] 3.3V

What is the expected current requirement for you supply pads (e.g., 0.1mA)

The average current on typical operation is 0.45 mA.
We preferred to use the 4mA supply pads

How many VSS Pads (0-2) *

1

How many IOVDD Pads (0-2) *

1

How many IOVSS Pads (0-2) *

1

How many Analog Pads (0-18) *

4

Have you completed the layout of what percentage of your full custom Design (choose from 0 = 0% to 10 = 100%) *

	1	2	3	4	5	6	7	8	9	10	
Nothing	○	○	○	○	○	○	○	○	○	○	Complete Layout

Progress Questionarie

Have you completed the layout of what percentage of your full custom Design (choose from 0 = 0% to 10 = 100%) *

	1	2	3	4	5	6	7	8	9	10	
Nothing	○	○	○	○	○	○	○	○	○	○	Complete Layout

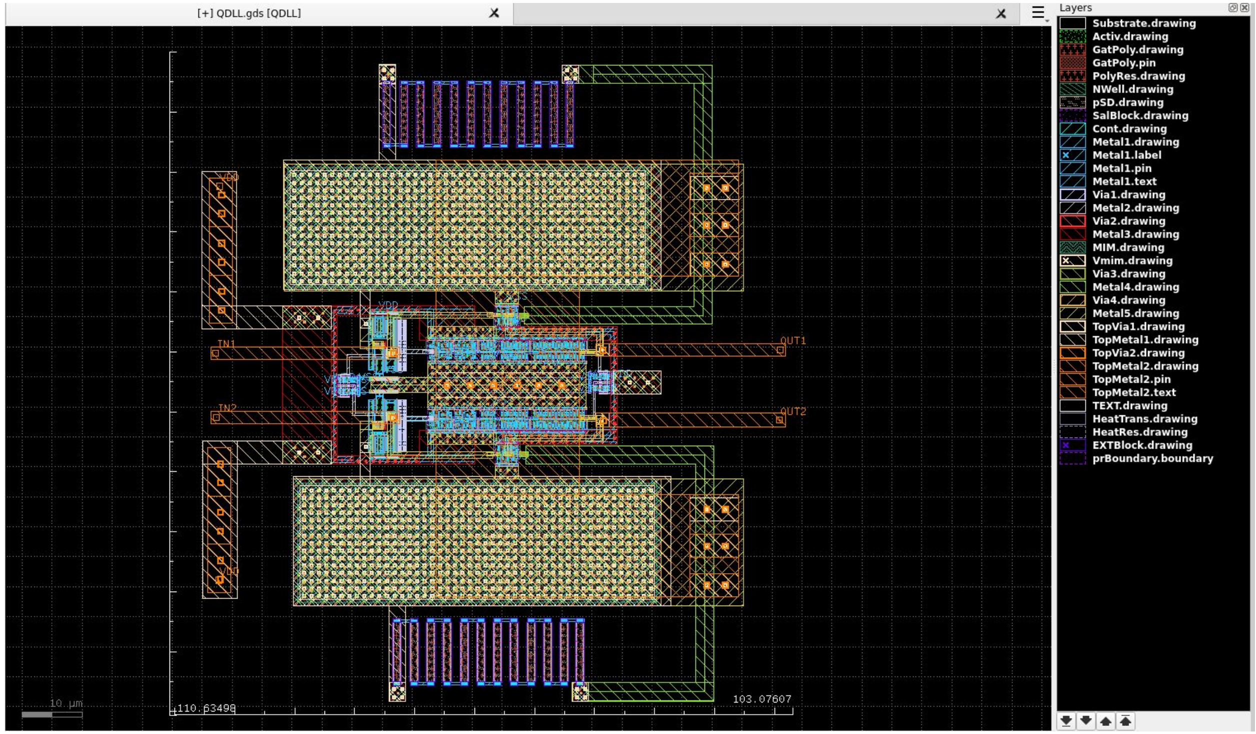
The digital blocks in your design, if needed, are they completed in RTL?

- ○ Yes
- ○ No
- ◉ Don't apply (fully analog)

Borrar la selección

What parts or blocks of your full custom Design are already laid-out and DRC & LVS verified with Klayout?

All blocks and the TOP have passed DRC and LVS verification with KLayout



Progress Questionarie

Write two sentences about the status of your Design and the design team familiarity with the Analog Open-source EDA flow.

Most members of the design team are graduate and undergraduate students with less than one year of experience using analog open-source EDA tools. Nevertheless, we have become well familiarized with the design flow, supported by the fact that two team members completed an open-source analog design course taught by Prof. Jorge Marín within the UNICCAS program, which has been very beneficial for transferring knowledge to the rest of the team.

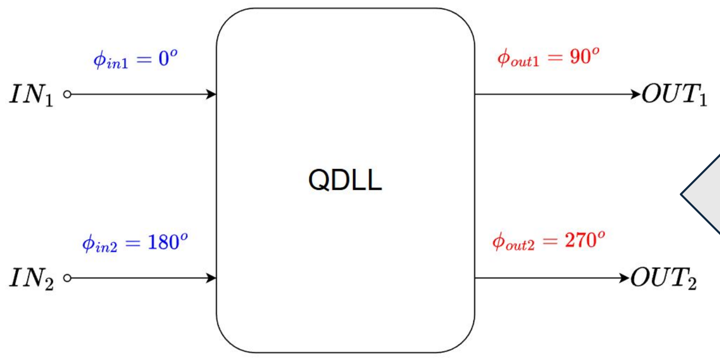
Write two sentences about difficulties, if any, the design team encountered with the Open-source tools and difficulties, PDK usage.

We encountered issues related to tool versions, as the UNICCAS repository does not provide the latest releases of Magic and KLayout. However, we have been able to work around these limitations and continue making progress in our project.

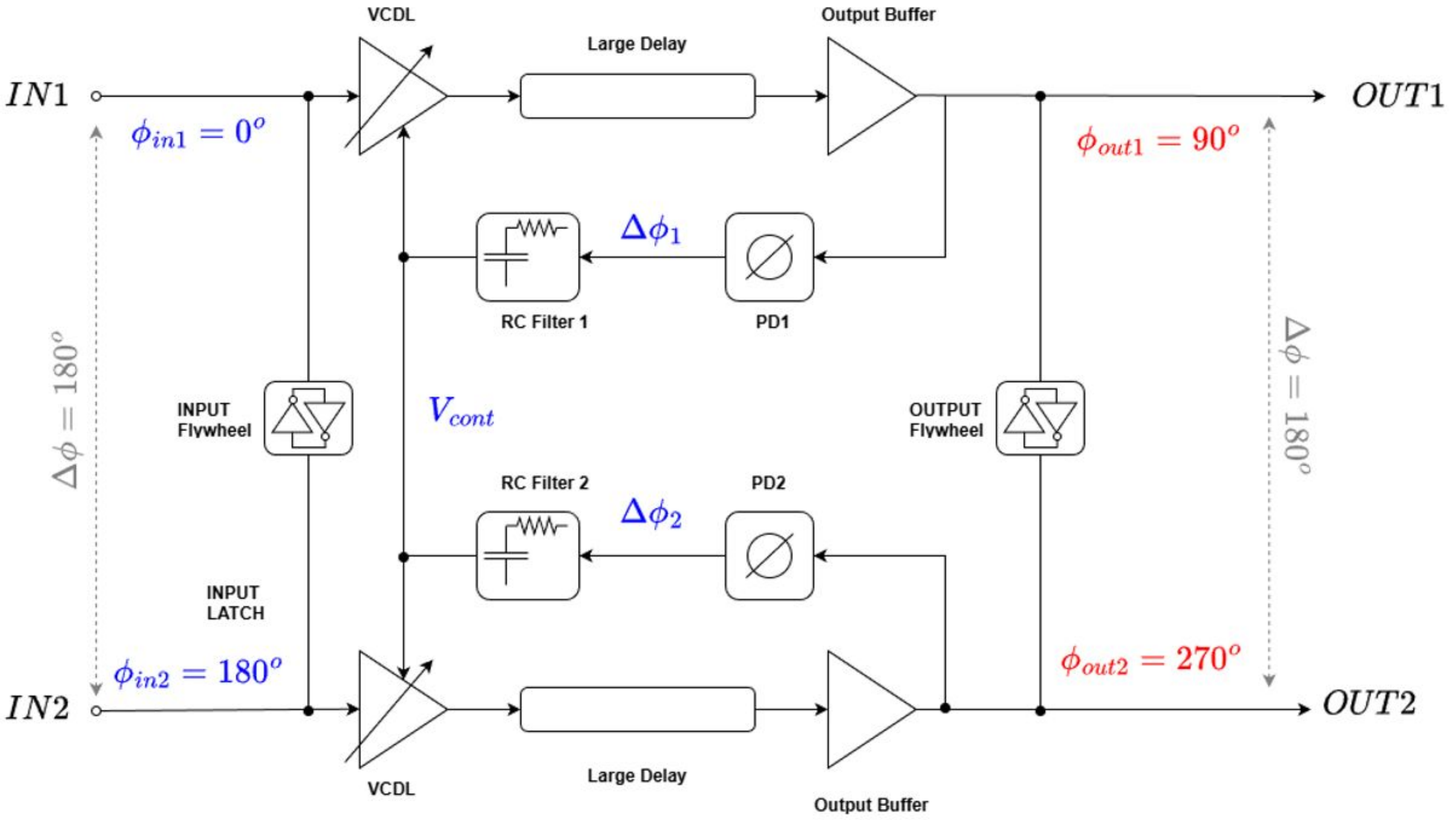
2- QDLL Project Diagram & Specifications

2 - QDLL Project Diagram

For this project, two single-ended DLLs with differential outputs are used to generate two differential signals with 90° and 270° phase shifts. The RC filters and the large VCDLs are designed to stabilize each branch at a **90°± 5°** .phase difference at **250 MHz**. To ensure synchronization between the signal pairs, flywheel cells are added.



QDLL block diagram



2 - QDLL Project Specifications

The design specifications were modified due to the high-frequency behavior of the I/O cells. As a result, the performance requirements were relaxed, along with adjustments to the project focus and the number of input/output pins.

Pins:

- 2 Bi-direcctional: VDD, VSS
- 2 inputs: inpad1, inpad2
- 2 outputs: outpad1, outpad2

Estimated Number of Pins:*

Input: *
2

Output: *
2

Bidirectional: *
2

ihp 130 nm

Design Type:* *

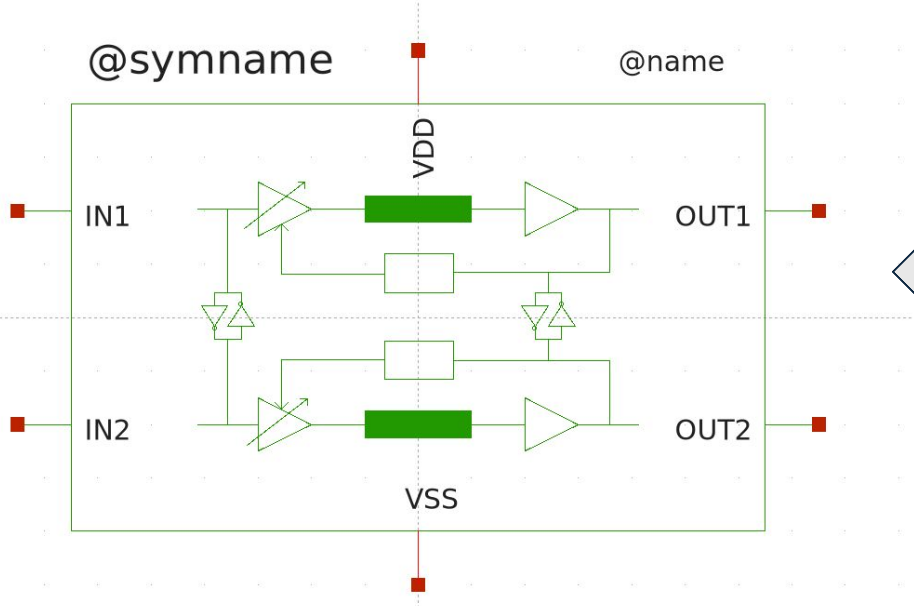
Analog

QDLL Specifications	
Boundary Conditions	**Requirement**
Techn.	CMOS
Node	130 nm
Supply Voltage	1.2 ± 5% V
Temperature	[0, 65, 125] °C
Input Amplitude	[0 , 1.2] V (CMOS)
Input Frequency	250±25 MHz
Input/Output Type	Fully Differential
Specifications	**Value**
Phase Difference	90 ± 5 degrees
Output Swing	[0 , 1.2] V (CMOS)
t_rise/t_fall output	<
Vcont_ripple	<
Output Jitter	-

3- QDLL Project Schematics & Layouts

3 - QDLL Schematics (TOP)

- Differential QDLL TOP:



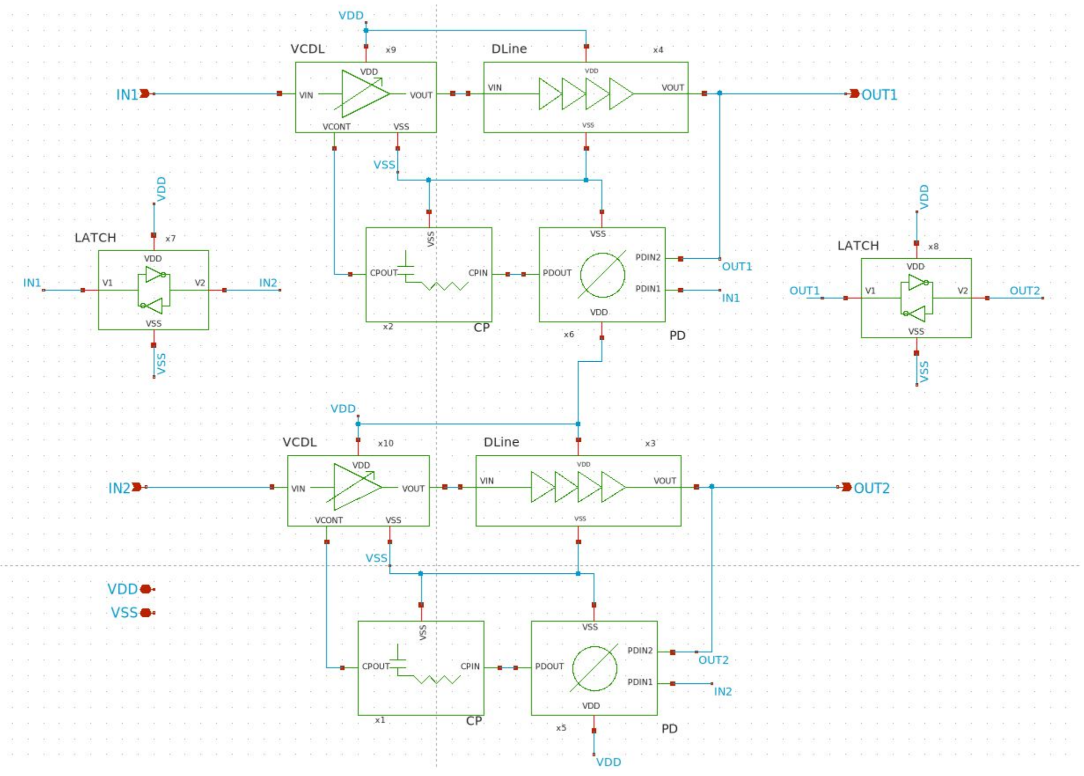
QDLL.sym

Estimated Number of Pins:*

Input: *
2

Output: *
2

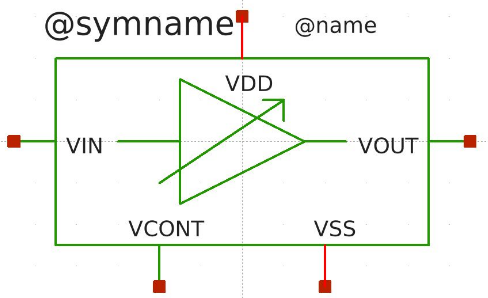
Bidirectional: *
2

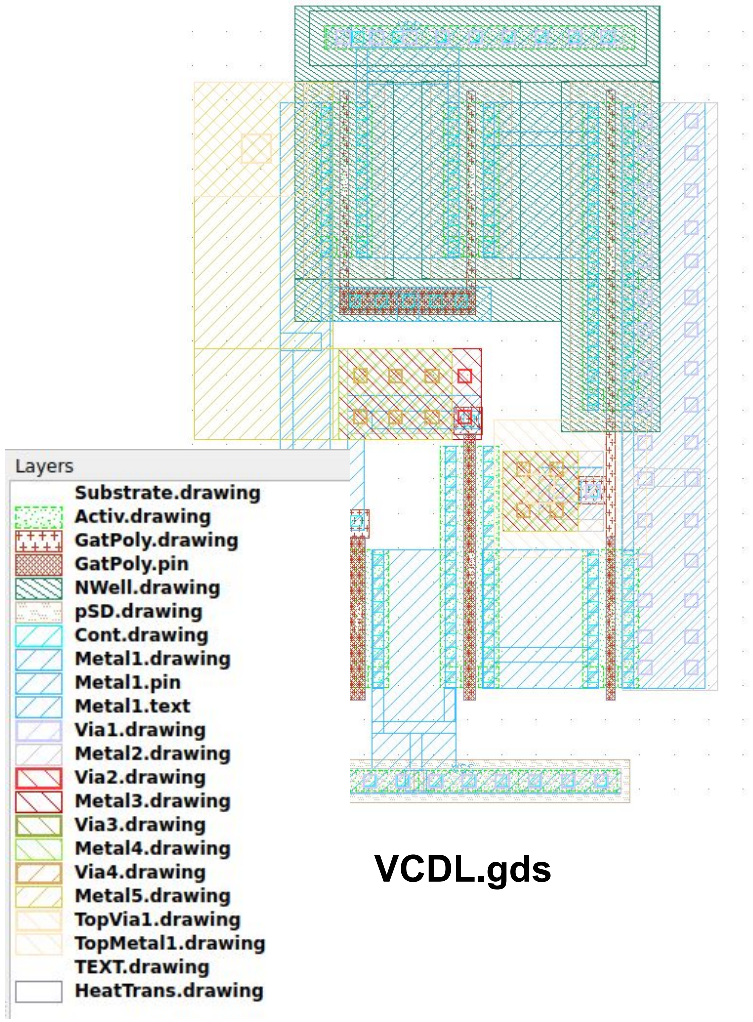


QDLL.sch

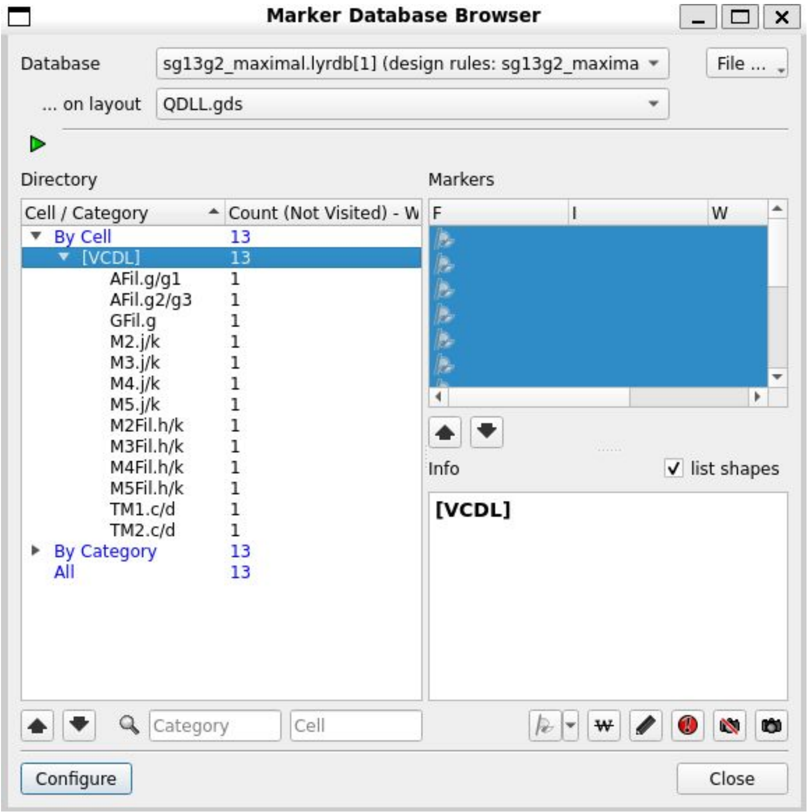
3 - QDLL Schematics & Layouts

- Voltage-Controlled Delay Line (VCDL) :

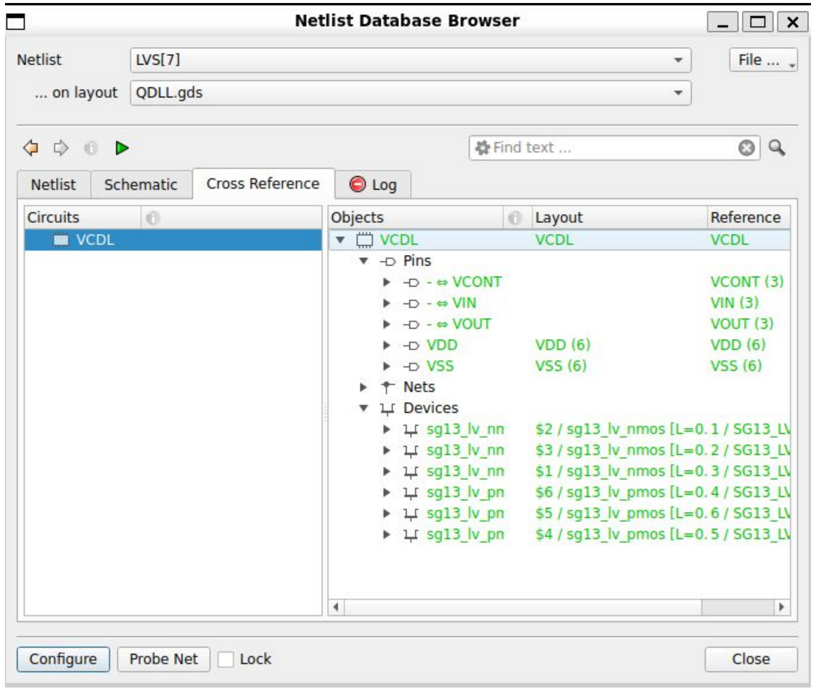




VCDL.gds



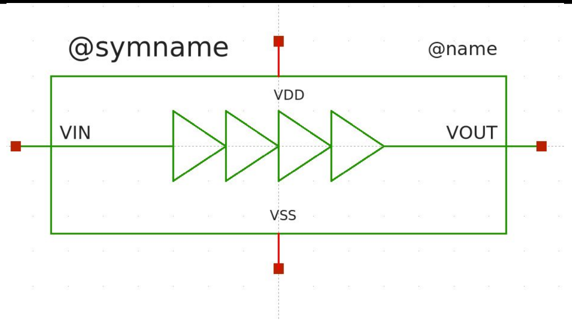
DRC clean (only density errors)



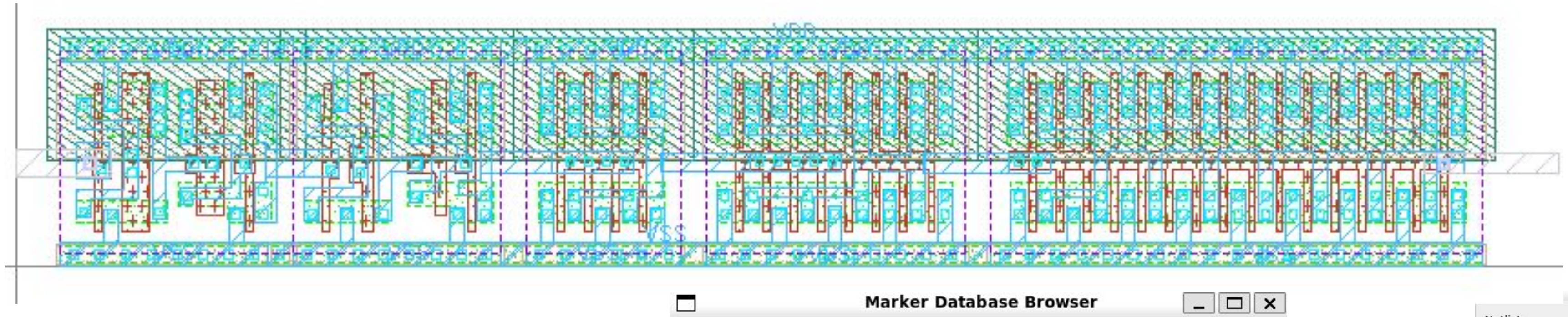
LVS Clean

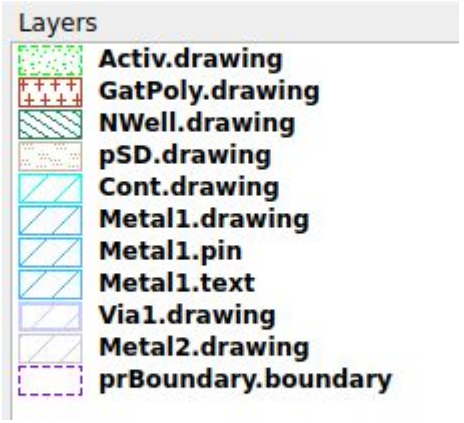
3 - QDLL Schematics & Layouts

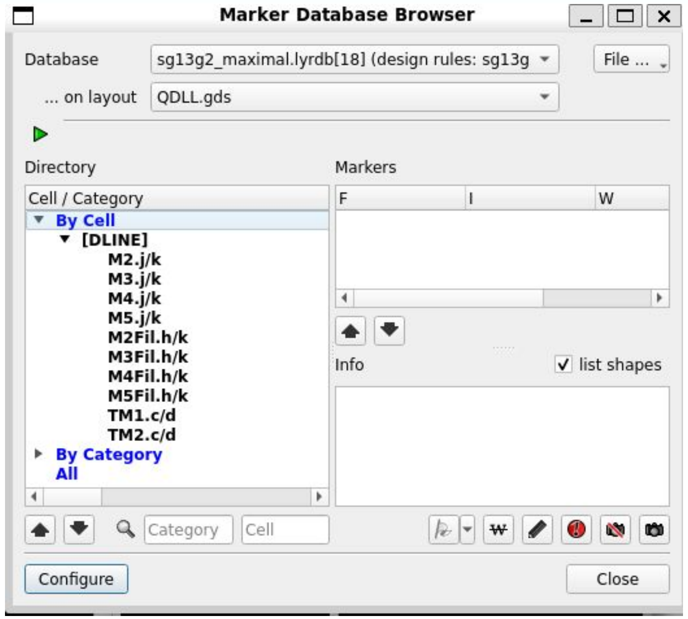
- Large delay Line + Buffer Stage (DLine) :

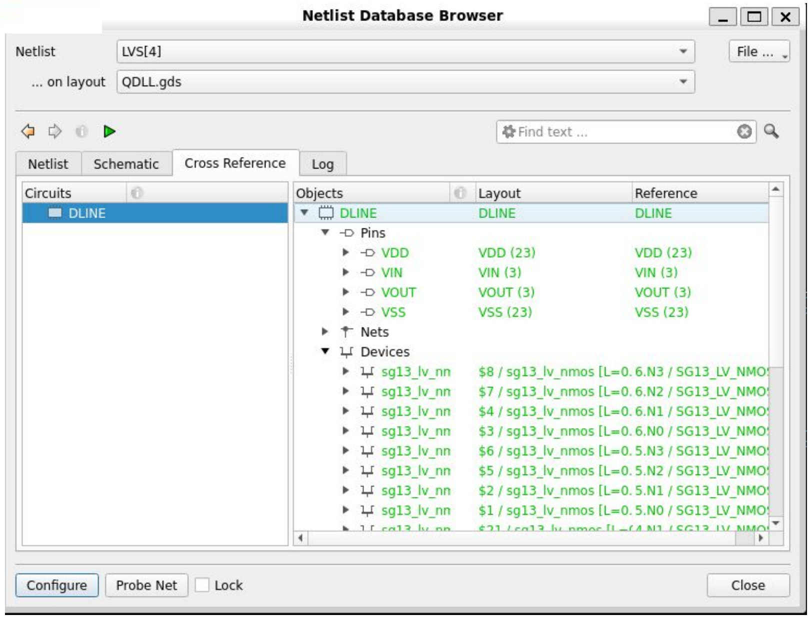


DLine.gds







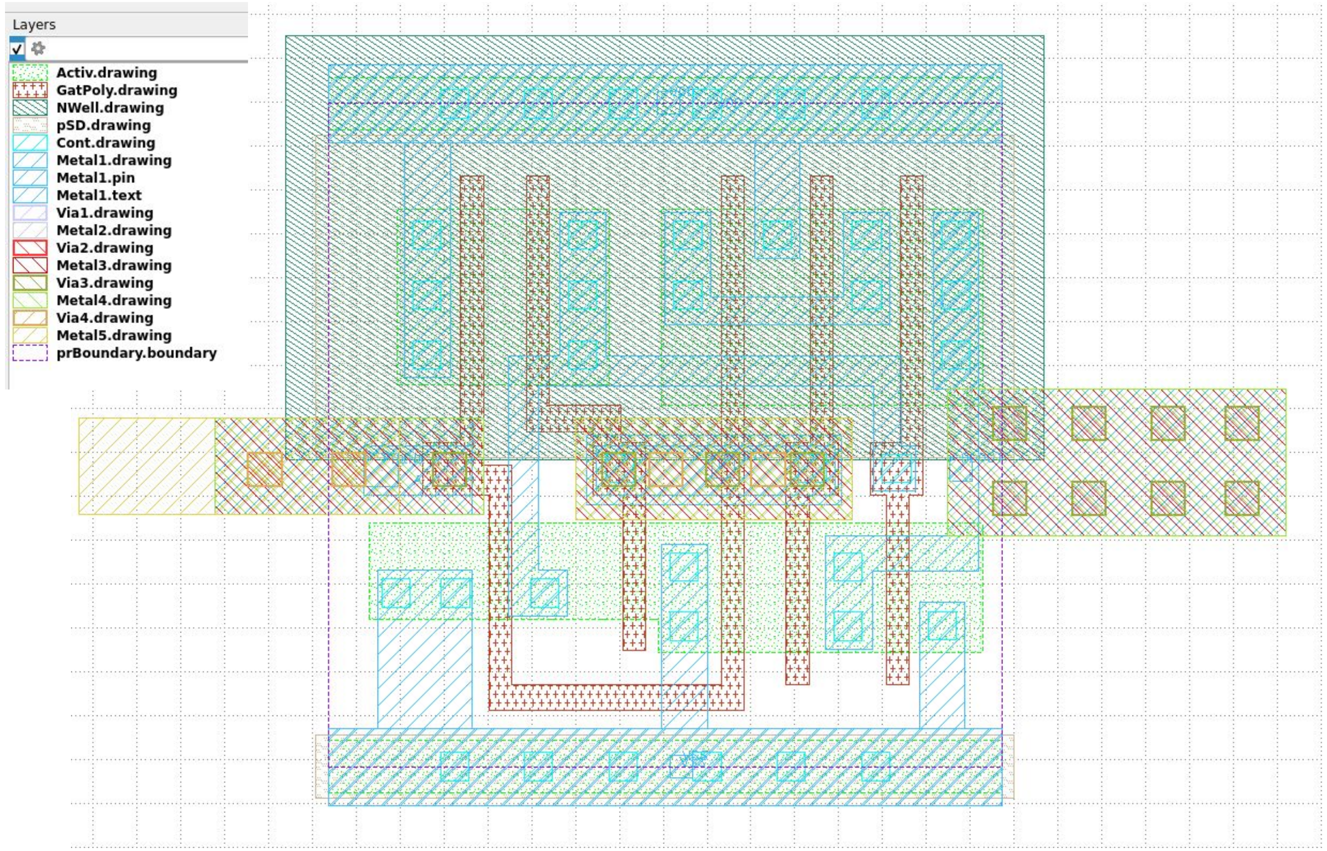


DRC clean (only density errors)

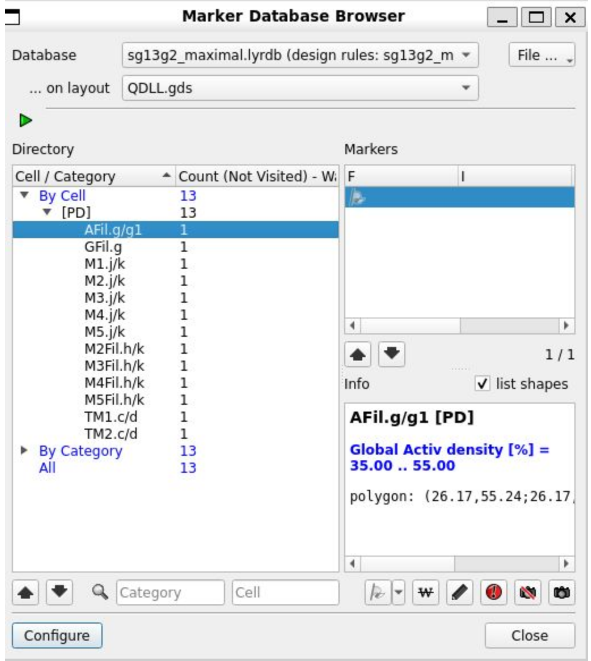
LVS Clean

3 - QDLL Schematics & Layouts

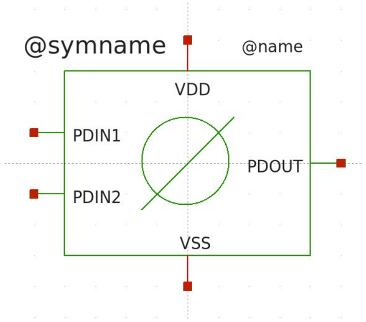
- Phase Detector (XOR cell) :



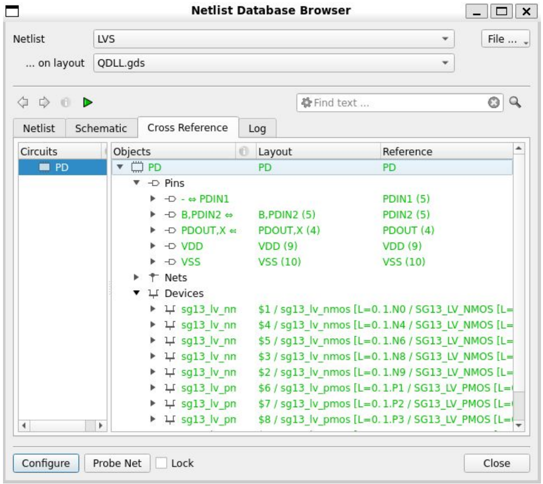
PD.gds



DRC clean (only density errors)



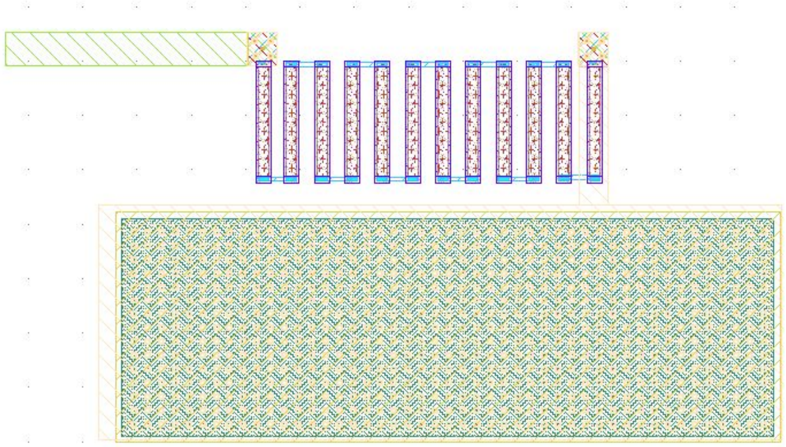
PD.sym

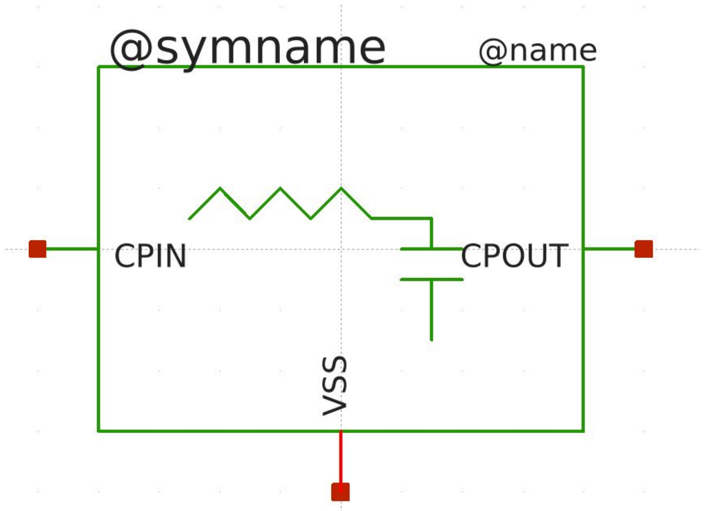


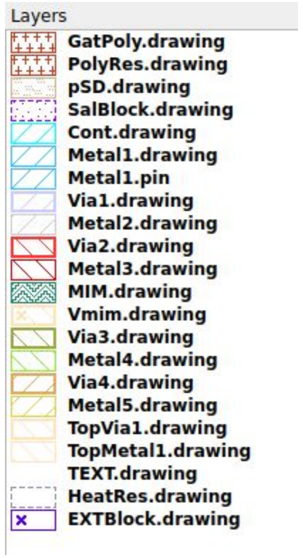
LVS Clean

3 - QDLL Schematics & Layouts

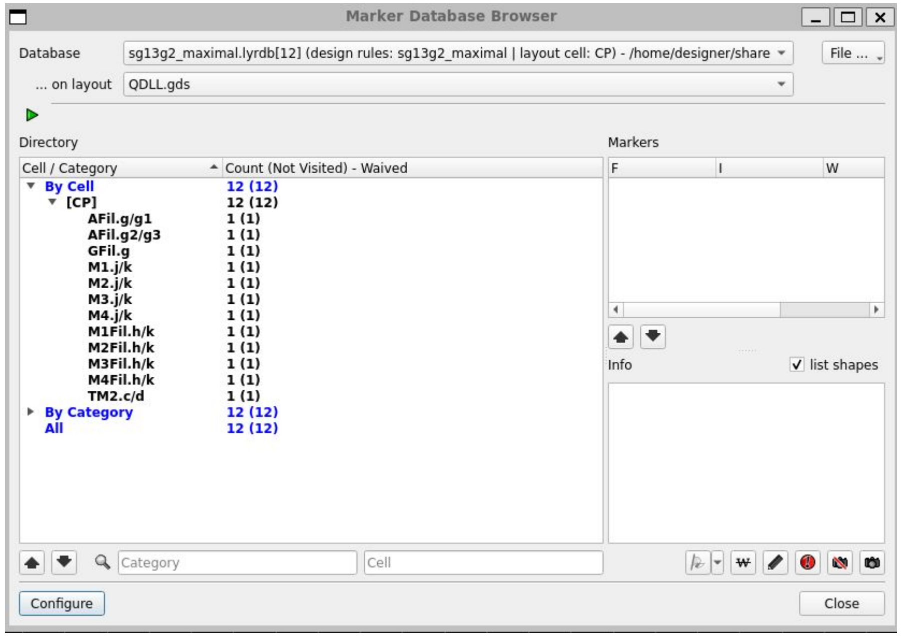
- Charge-Pump/ RC Loop Filter:

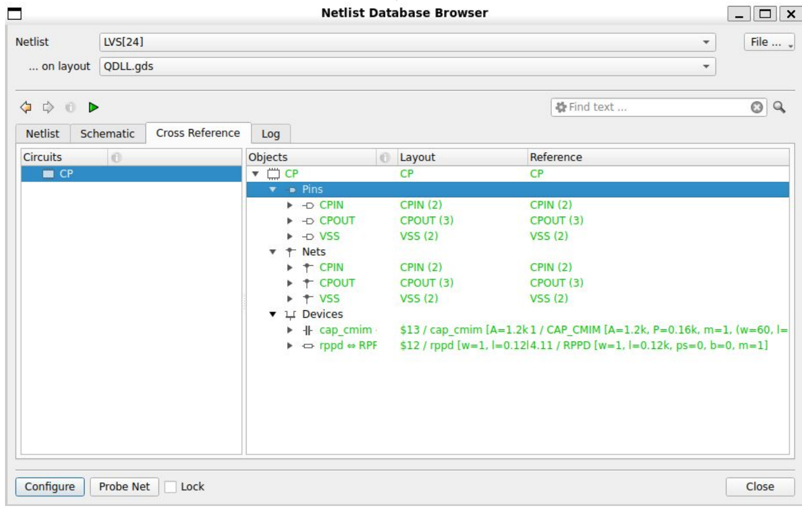






CP.gds



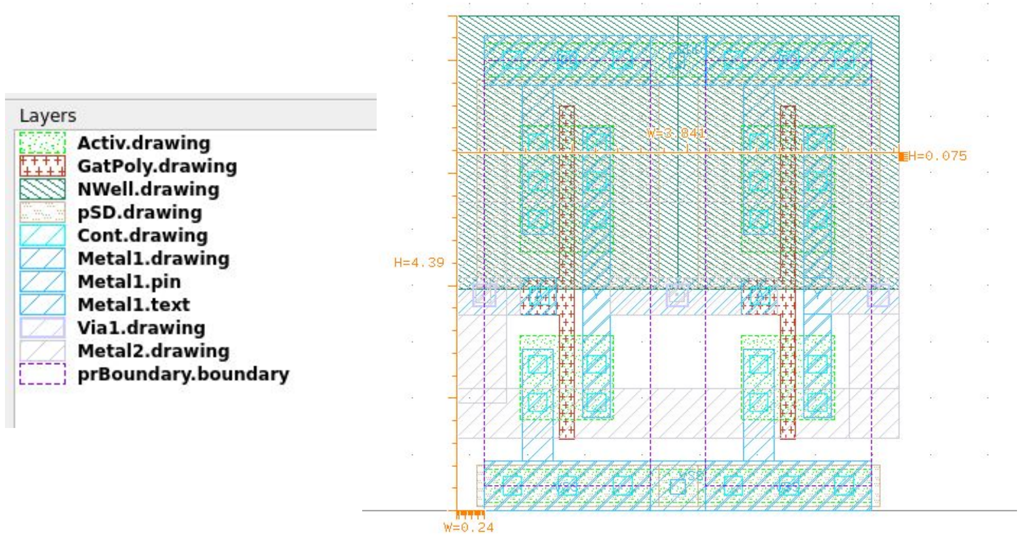


DRC clean (only density errors)

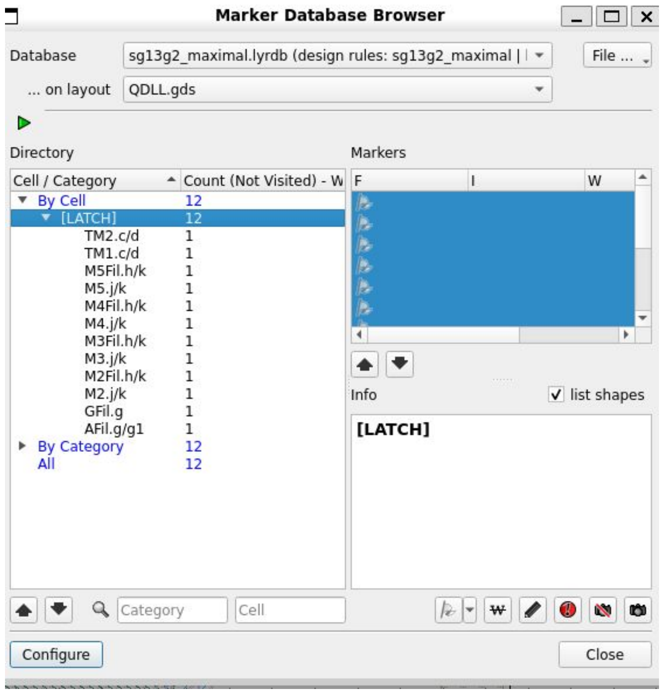
LVS Clean(depends on !sub connection)

3 - QDLL Schematics

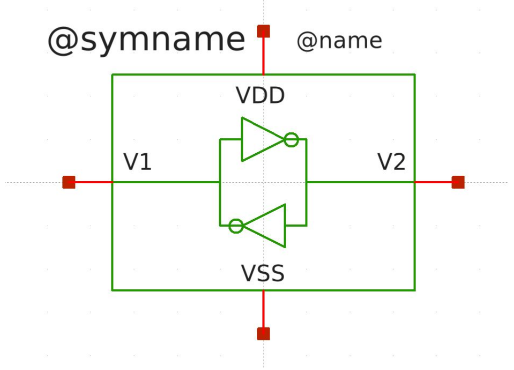
- Flywheel Latchs:



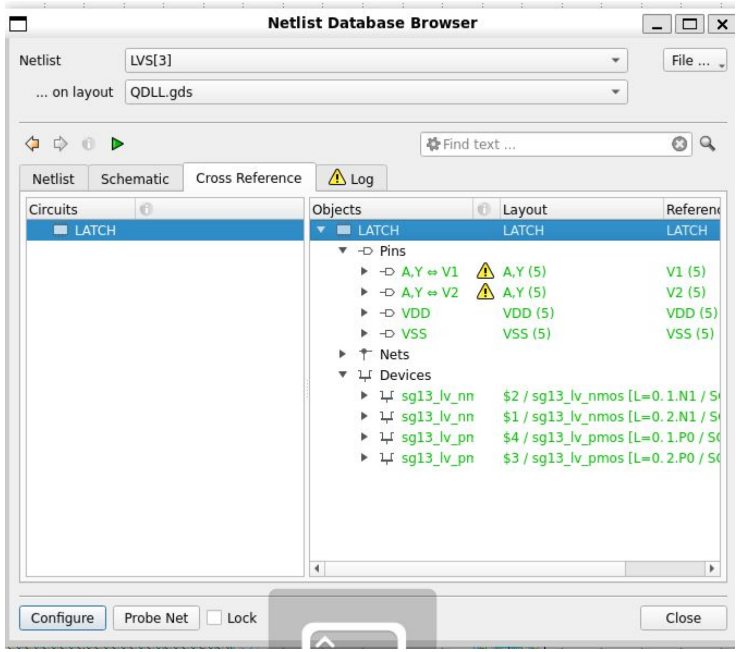
LATCH.gds



DRC clean (only density errors)



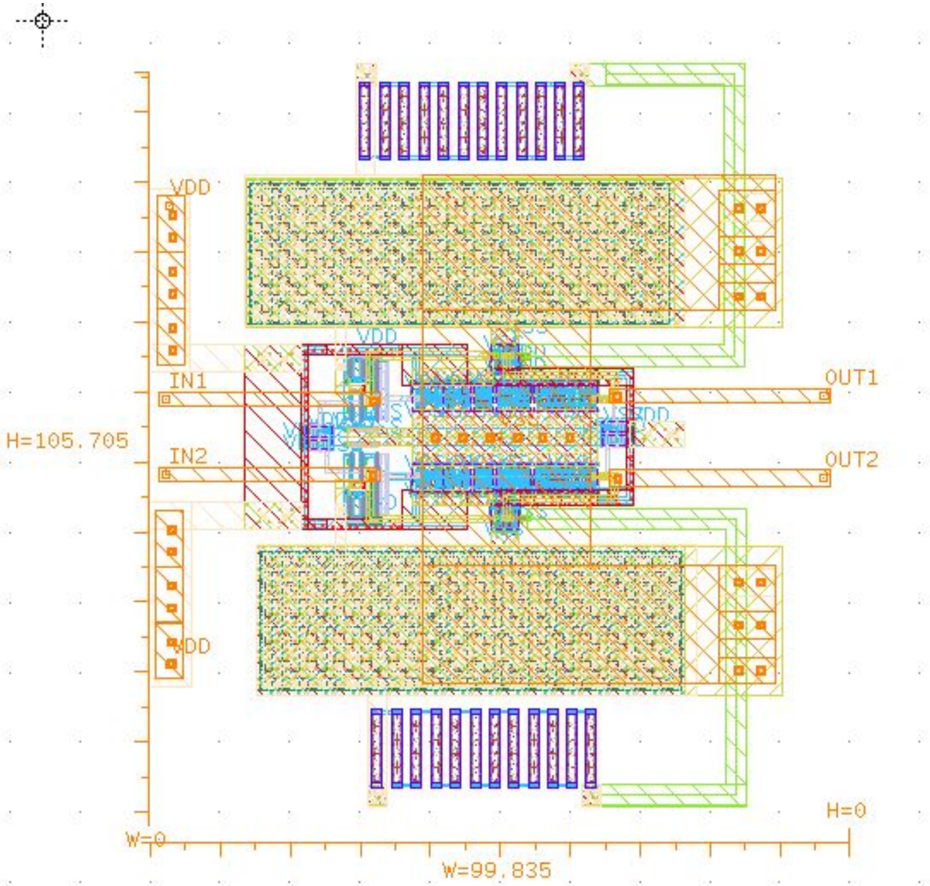
LATCH.sym

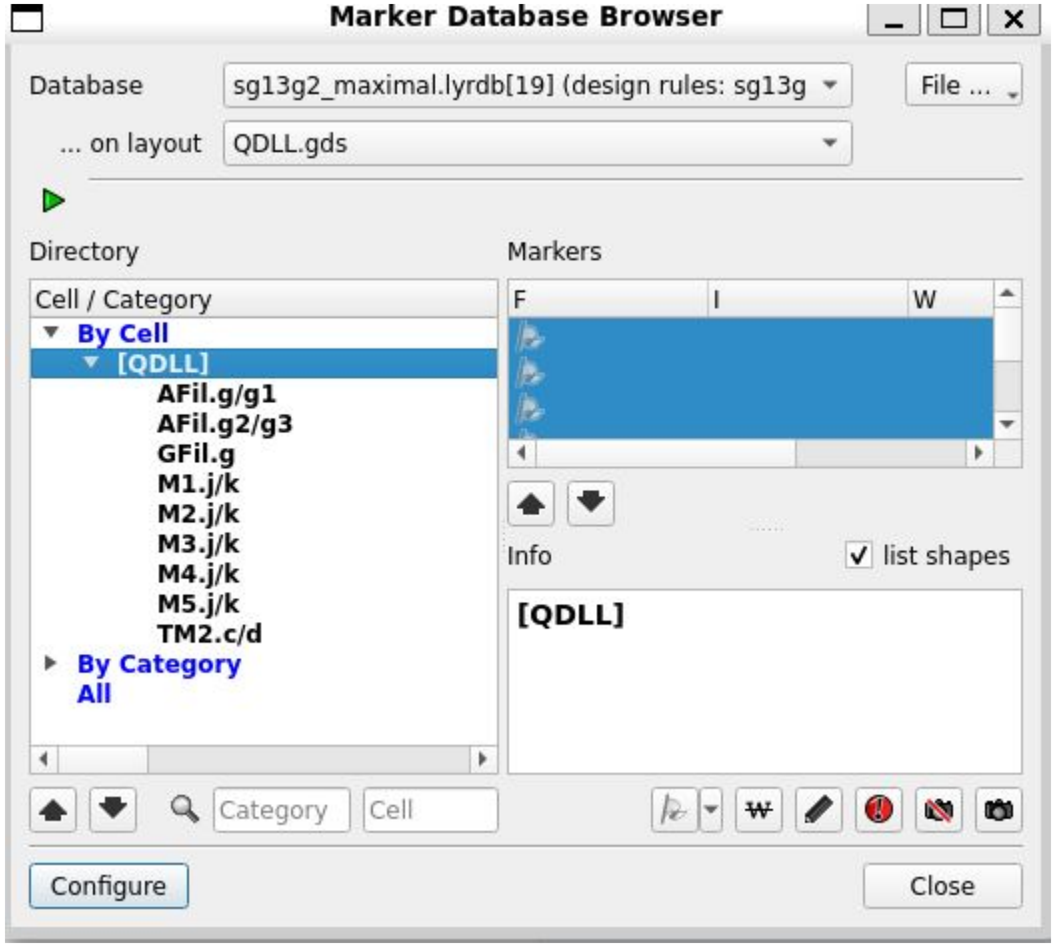


LVS Clean

4- QDLL TOP Layout

QDLL.gds Top Layout Verification



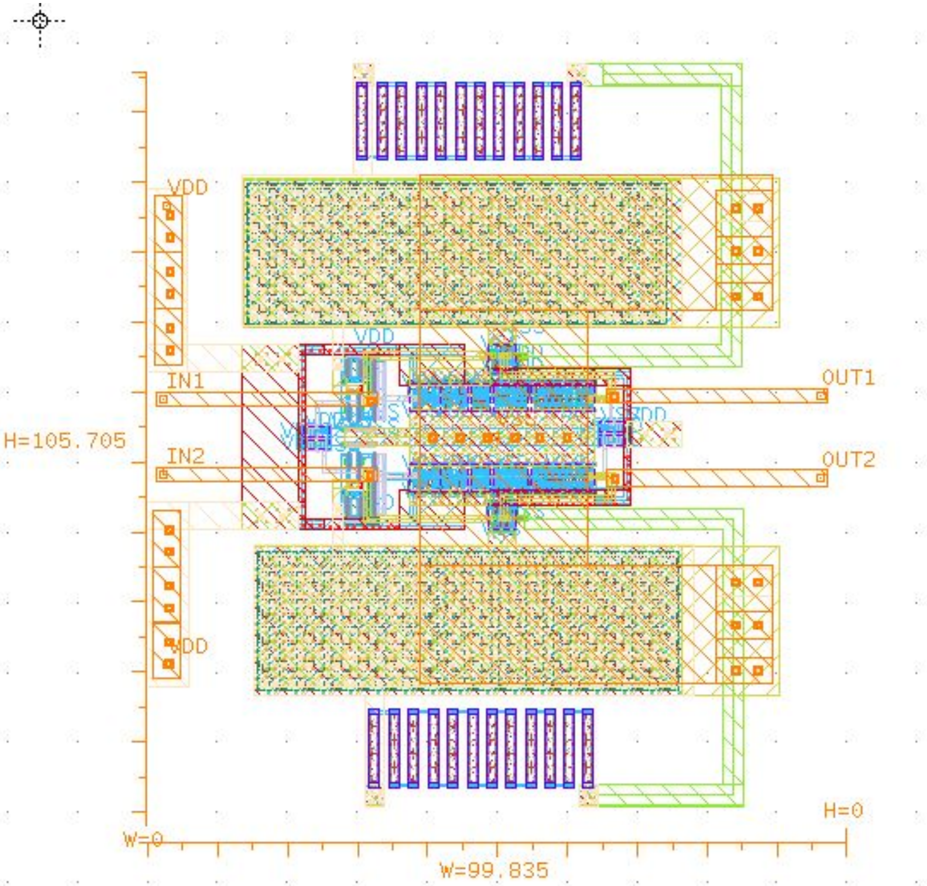


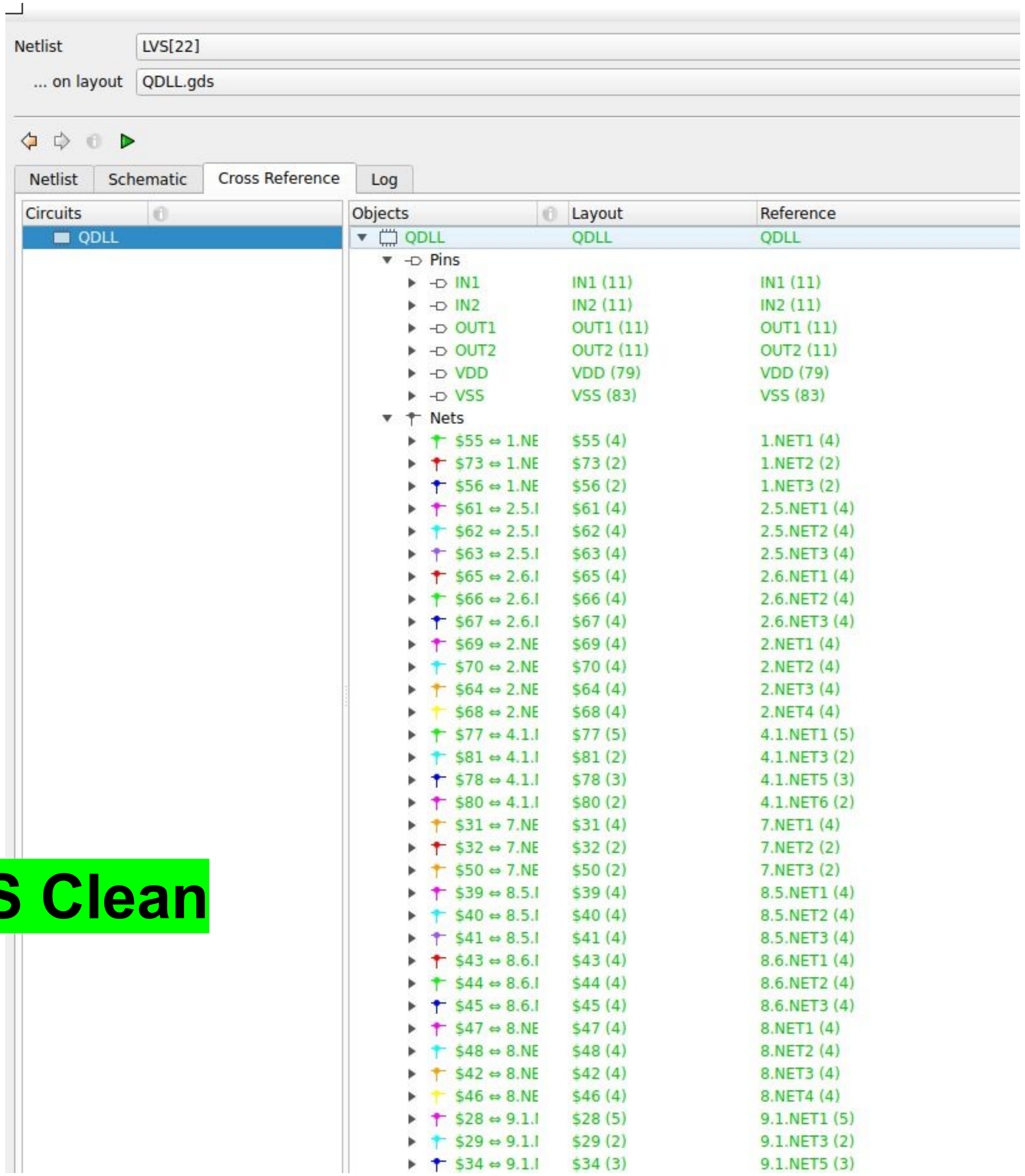
Vertical View

Area aprox: 100um x 100 um = 0.1um^2 =

DRC Clean (ONLY DENSITY ERRORS, SOLVING ON INTEGRATION STAGE

QDLL.gds Top Layout Verification



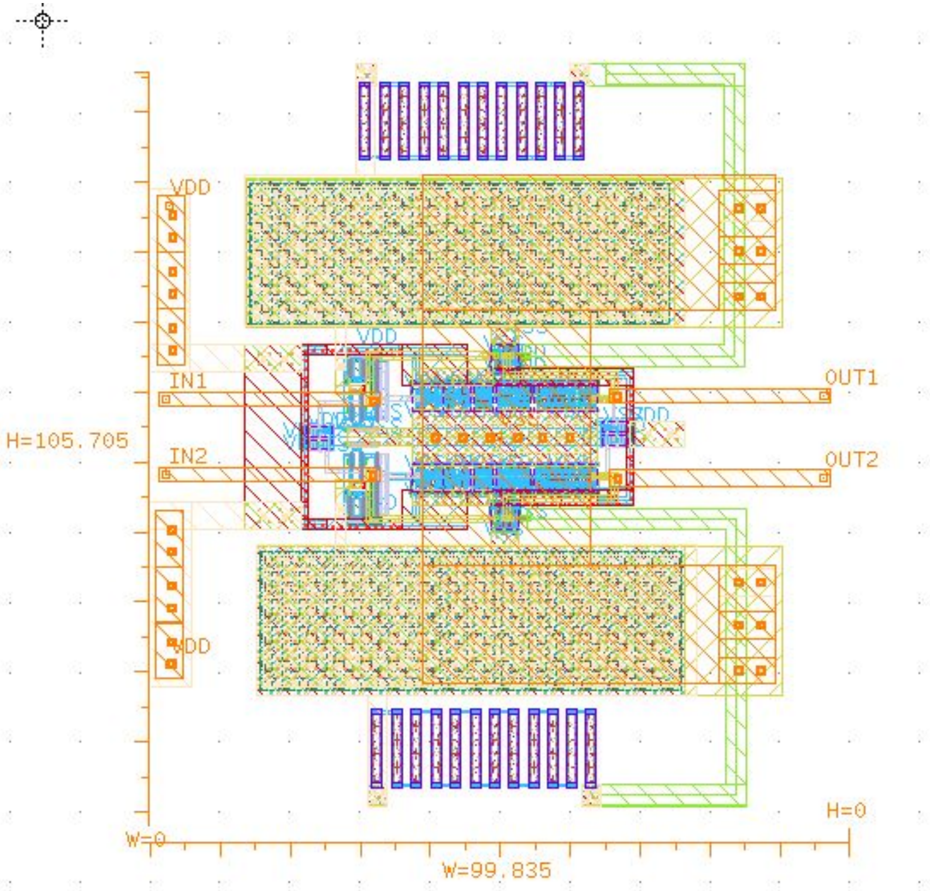


LVS Clean

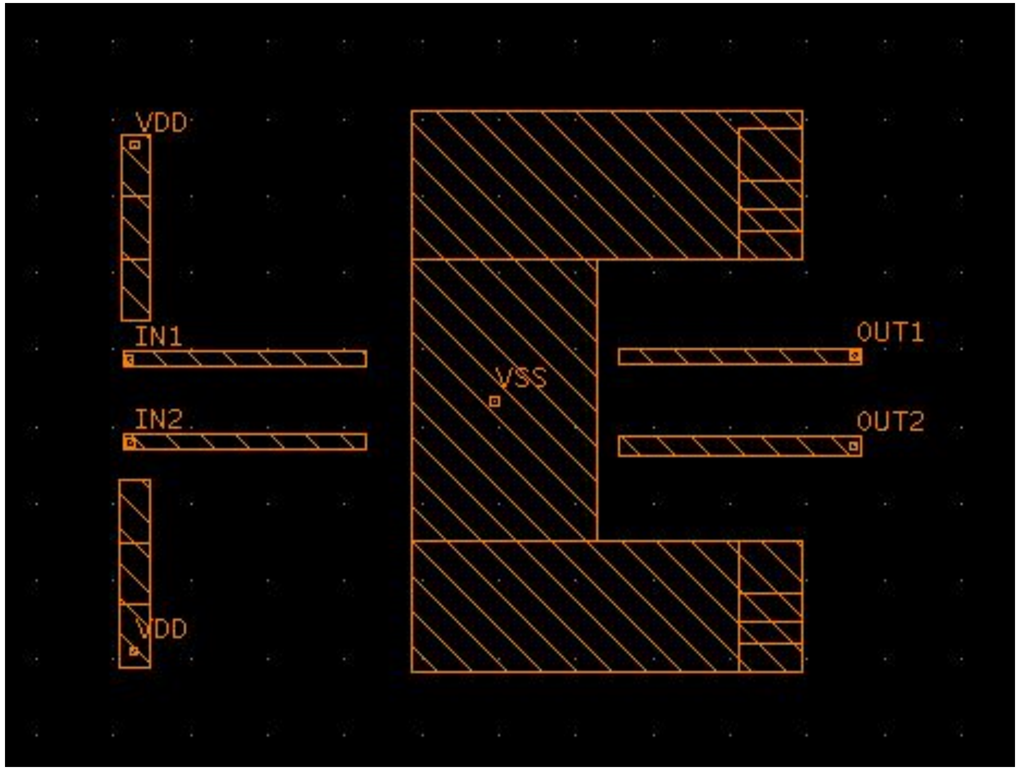
Vertical View

Area aprox: 100um x 100 um = 0.1um^2 =

QDLL.gds Top Layout CONNETION



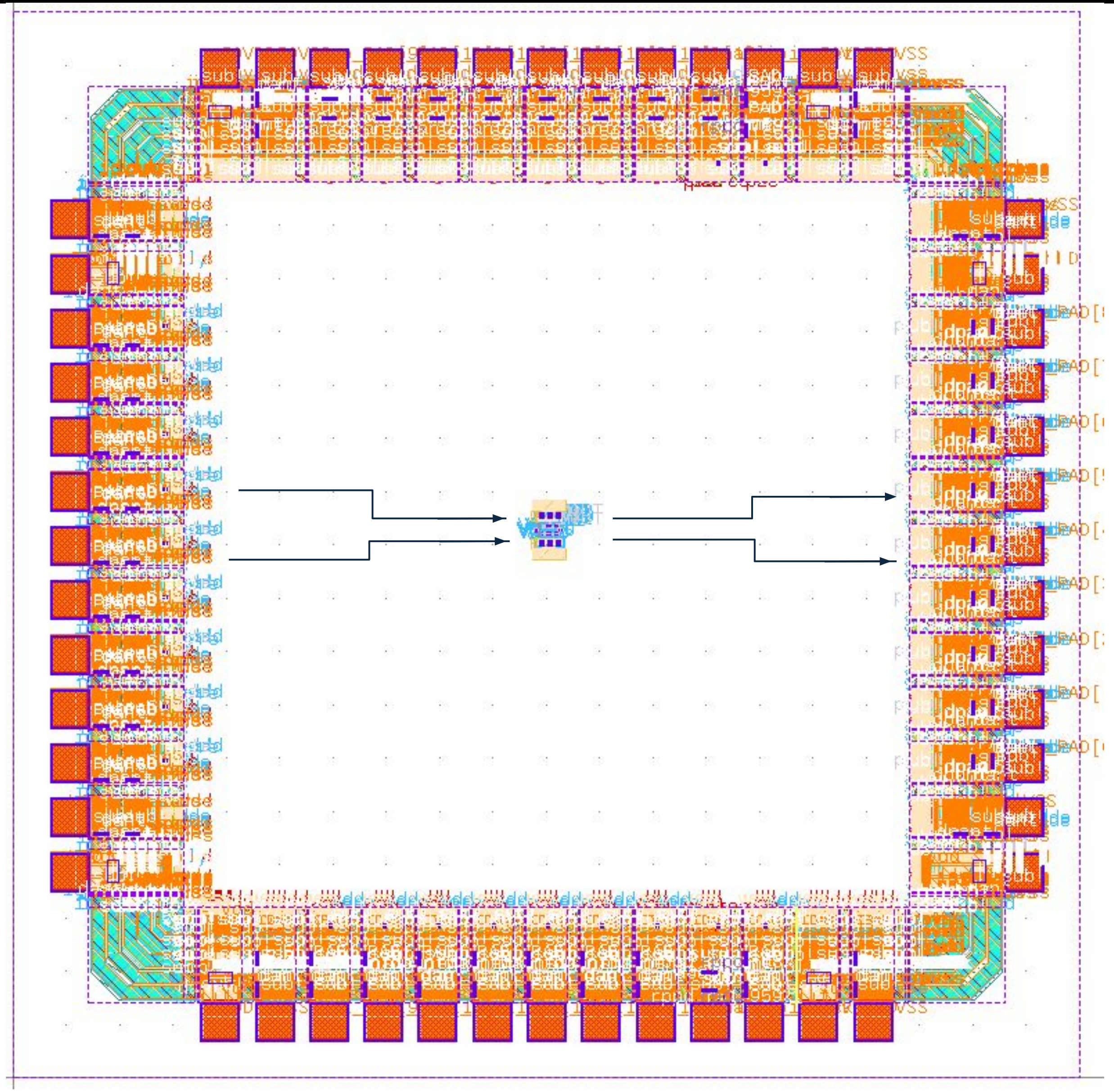
Vertical View
Area aprox: 100um x 100 um = 0.1um^2 =



TOP METAL 2 CONNECTIONS:

- **POWER:** VDD, VSS
- **INPUTS:** IN1, IN2
- **OUTPUT:** OUT1, OUT2

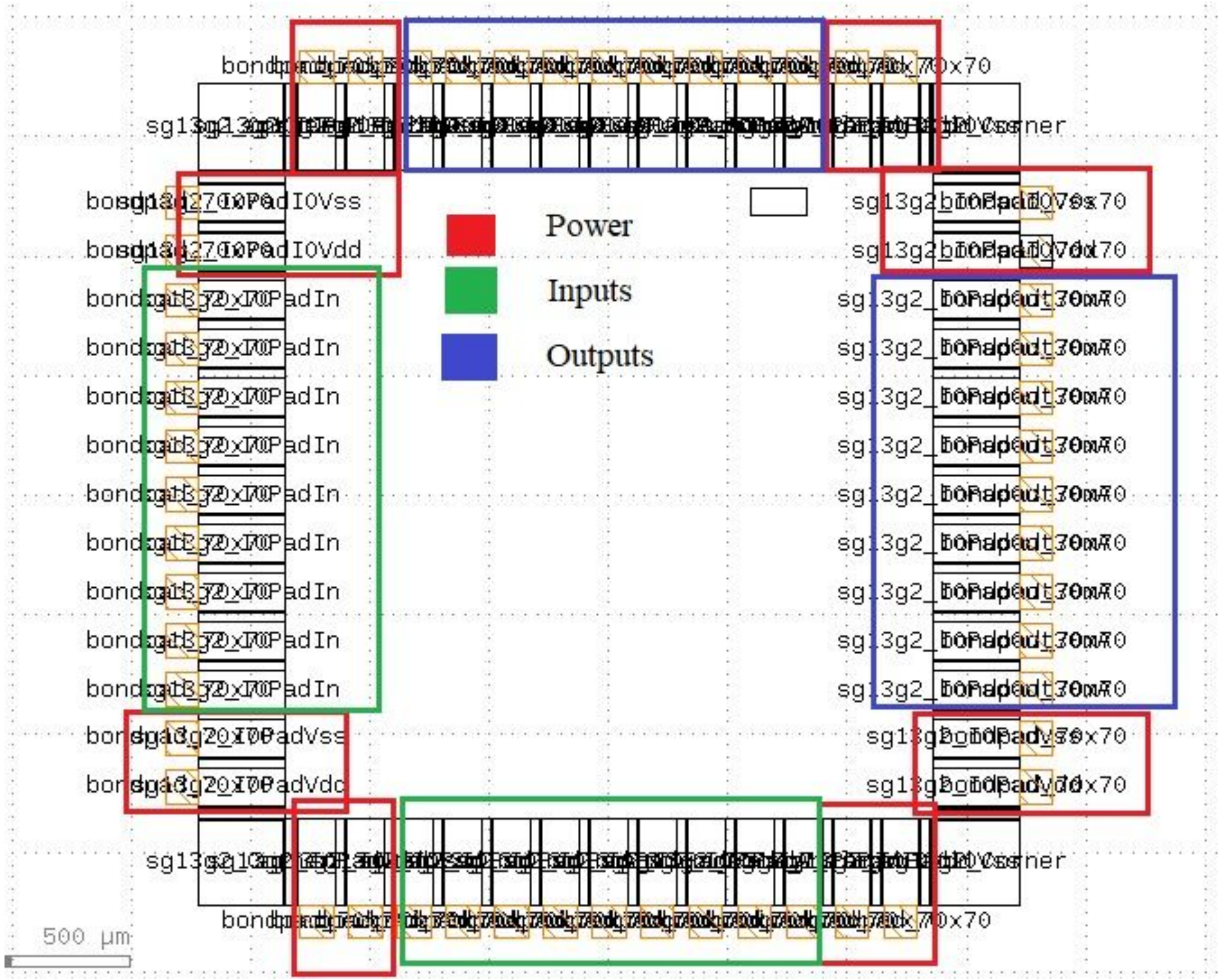
QDLL Top Layout Placement v1



One possible goal would be to place the QDLL vertically and connect the inputs on the left and the outputs on the right with high metal top routing.

For a 250 MHz signal, standard IO cells have been tested and found to perform well. However, if available, we prefer to interface with 50 Ω systems using IO cells with matched impedance.

QDLL layout within the pad ring



5 - QDLL Pins and IO CELLS

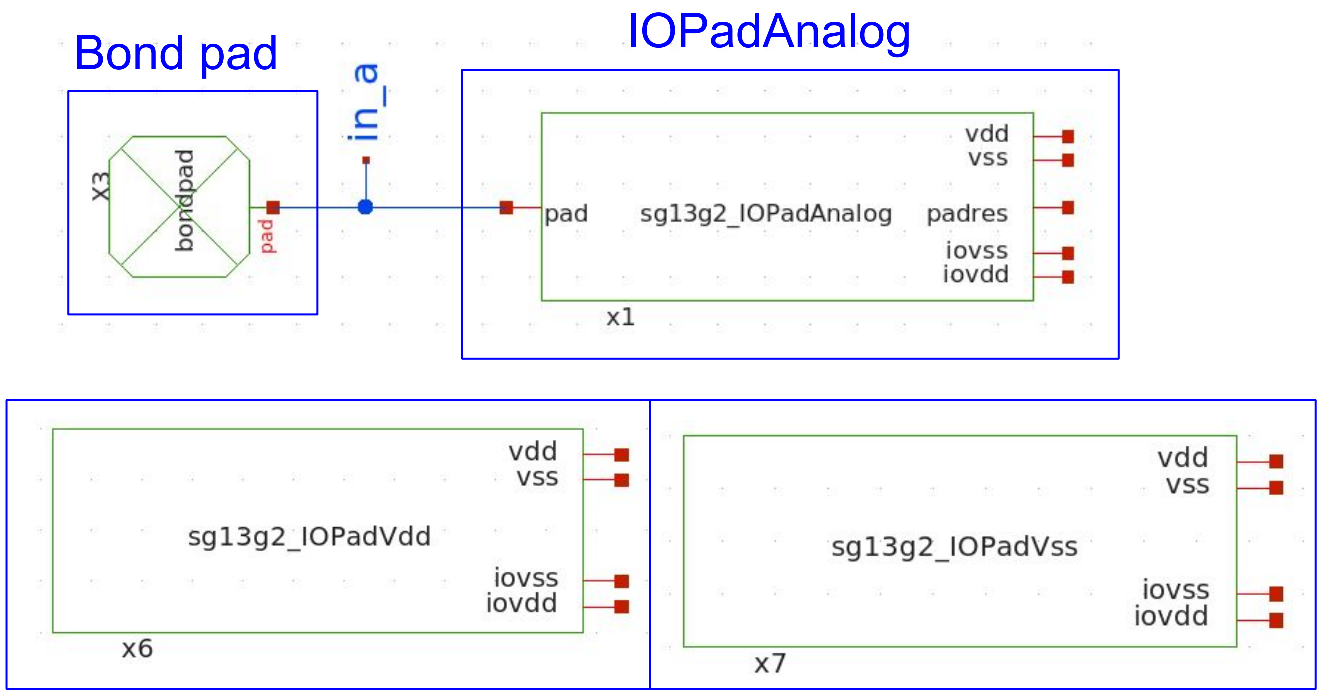
Pins and IO Cells

QDLL Pins

<u>N° Pins:</u> 4

- 2 Input Pins
- 2 Output Pin
- 2 Power Supply Pins (VDD , VSS)

Pin Type	Name	IOCELL
Input	inpad1	sg13g2_IOPADAnalog
Input	inpad2	sg13g2_IOPADAnalog
Output	outpad1	sg13g2_IOPADAnalog
Output	outpad1	sg13g2_IOPADAnalog
Power Supply	VDD	sg13g2_IOPadVdd
Power Supply	VSS	sg13g2_IOPadVss



Power Supply IOPads

6- QDLL Project Status

Project Status

Project Status by Block:

TOP	Sub-top	Sub-Block	Symbols & Schematics	Testbench	Prelayout results	IOCELLS results (TT)	IOCELLS results (SS-FF)	Layout (DRC + LVS)	Postlayout results
QDLL TOP	-	-	Done	Done	Done	Done	W/P	Done	W/P
-	SE_QDLL	-	Done	Done	Done	Done	WIP	Done	WIP
-	-	VCDL	Done	Done	Done	Done	W/P	Done	W/P
-	-	DLine	Done	Done	Done	Done	WIP	Done	WIP
-	-	CP	Done	Done	Done	Done	WIP	Done	WIP
-	-	PD	Done	Done	Done	Done	WIP	Done	WIP
-	LATCH	-	Done	Done	Done	Done	WIP	Done	WIP

For this delivering, we were working on the DRC (minimum and maximum) verification of all cells to achieve the desired LVS verification. Next, we plan to compare the extracted views using Magic in order to obtain more realistic results.

On the other hand, multi-corner simulation results are desirable for our project, and we are actively working toward that goal.